Course: BTech

Semester: 3

Prerequisite: Basic knowledge of programming

Rationale: The practical component reinforces Core Java concepts through hands-on programming. It helps students apply OOP, collections, exception handling, file I/O, multithreading, and Java 8 features to solve real-world problems, preparing them for advanced development and industry roles.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
					Internal Marks			External Marks		
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	T	CE	P	T	P	
0	0	2	0	1	-	-	20	-	30	50

SEE - Semester End Examination, **T** - Theory, **P** - Practical

List of Practicals	
Sr.	**Topics**
1.	Write a Java program to display "Hello World" and demonstrate the Java program structure and demonstrate use of variables, data types, and type casting.
2.	Write a Java program to perform arithmetic, relational, and logical operations, bitwise and other operators.
3.	Write a Java program using below conditional statements. a) Even or Odd using if.. else statement. b) Roots of Quadratic Equation using else if ladder. c) Largest of three numbers using nested if else. d) Find out the week day using switch statement.
4.	Write a Java program to demonstrate looping constructs: a) Reverse of a number using while loop. b) Prime number using do while loop. c) nth term of fibonacci sequence using for loop.
5.	Write a Java program to perform methods and example on recursion.
6.	Write a Java program to perform operations on one-dimensional and multi-dimensional arrays.
7.	Write a Java program to demonstrate 1D - Arrays & 2-D Arrays. a) Maximum value and Second Maximum value without duplicates. b) Sort the names in Ascending Order. c) Addition of two matrix. d) 3x3 Matrix Multiplication.
8.	Write a Java program to demonstrate string handling using String, StringBuffer, and StringBuilder.
9.	Write a Java program to check the word is palindrome or not.
10.	Write a Java program to create a class and object and demonstrate constructors.
11.	Write a Java program to demonstrate encapsulation using access specifiers.
12.	Write a Java program to demonstrate inheritance: a) Single Inheritance. b) MultiLevel Inheritance. c) Hierarchial Inheritance. d) Hybrid Inheritance.
13.	Write a Java program to demonstrate use of the **this** and **super** keywords.

14.	Write a Java program to demonstrate polymorphism (**method overloading and overriding**).
15.	Write a Java program to implement abstraction using abstract classes and interfaces(Multiple Inheritance).
16.	Write a Java program to demonstrate exception handling using try-catch-finally blocks.
17.	Write a Java program to create and use custom exceptions.
18.	Write a Java program to implement List interface using ArrayList and LinkedList.
19.	Write a Java program to implement Set interface using HashSet and TreeSet.
20.	Write a Java program to implement Queue interface using Priority Queue and Deque.
21.	Write a Java program to implement map interface using HashMap and LinkedHashMap.
22.	Write a Java program to demonstrate multithreading using Thread class and Runnable interface.

Course Outcome

After Learning the Course the students shall be able to:	
CO	**Description**
CO1	Write and execute Java programs using proper syntax and coding standards.
CO2	Design object-oriented solutions using classes, interfaces, inheritance, and packages.
CO3	Implement collections, arrays, strings, and custom exception handling in programs.
CO4	Perform file handling and object serialization using Java I/O streams.
CO5	Develop multithreaded and functional-style programs using Java 8 features.
CO6	Develop database-driven Java applications using JDBC, DAO pattern, and layered architecture.